Haijun Gong, PhD

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Research Interests

Systems Biology/Bioinformatics	Statistics/Data Science
Machine Learning	Deep Learning
Quantum Computing	Model Checking/Formal Verification

Professional Appointment

07/2025 –	Professor of Statistics (Tenured) Department of Mathematics and Statistics Saint Louis University, USA
07/2017 – 06/2025	Associate Professor of Statistics (Tenured) Saint Louis University, USA
08/2018 – 08/2019	Associate Professor of Statistics (on sabbatical) Research School of Finance, Actuarial Studies and Statistics Australian National University, Australian
08/2012 – 06/2017	Assistant Professor of Statistics Saint Louis University
09/2009 – 08/2012	Postdoctoral Fellow Computer Science Department Advisor: Edmund Clarke (Turing Awardee-2007) Carnegie Mellon University (CMU), Pittsburgh, PA

Education

08/2009	Ph.D., Department of Physics, Carnegie Mellon University, USA Advisor: Russell Schwartz (Head of Computational Biology Department)
05/2008	M.S., Department of Statistics, Carnegie Mellon University

Citizenship

- Chinese
- USA Permanent Resident

Funded Projects

- NIH-NIGMS R15 09/2022-08/2026
  Novel Systems Biology Methods for the Cell-type-specific Regulatory Networks Reconstruction from scRNA-seq Data.
  1R15GM148915-01: $454,500; 3R15GM148915-01S1: $90,096
  Total Amount: $544,596
  Role: **PI**

Completed Projects

- President's Research Funds (PRF), Saint Louis University 05/2022-01/2024
  GAN-based deep learning methods for the scRNA-sequencing data analysis
  Amount: $25,000
  Role: **PI**

- NIH-NIGMS R15 09/2018-08/2022
  A Systems Biology Approach to Investigate the Structure Changes of Biological Network
  Amount: $454,500
  Role: **PI**
- NIH R01 Tomasz Heyduk (PI) 01/2015-06/2017
  New Bioanalytical Methods Based on Next Generation Sequencing
  Role: **co-Investigator**
- President's Research Funds (PRF), Saint Louis University 03/2014-08/2015
  Computational Analysis of Signaling Pathways in the Cancer
  Amount: $25,000
  Role: **PI**

Publication

*Note: underline with * denotes SLU undergraduates, ** denotes SLU graduates under my direction*

A1. Tong Si, **Haijun Gong**, Challenges and Advances in Bioinformatics and Computational Biology. *Current Issues in Molecular Biology* 7(2), 48(2), 185 (2026)

A2. Graham Bishop**, Tong Si, **Haijun Gong**, Blue Noise-based Generative Models for the Imputation of Time-series Data. *15th International Conference on Information Science and Technology* (2025)

A3. Graham Bishop**, Tong Si, **Haijun Gong**, Imputation of time-series scRNA-seq data with a time-varying blue noise diffusion model. *Briefings in Bioinformatics* (2025)

A4. Lingling Zhang, Tong Si, Lucas Koch*, **Haijun Gong**, Gene Regulatory Network Inference from Pseudotime-Ordered scRNA-seq Data via Time-Lagged Divergence Measures. Proceedings of 12th International Conference on Bioinformatics Research and Applications (ICBRA) (2025)

A5. Graham Bishop*, Tong Si, Isabelle Luebbert*, Noor Al-Hammadi, **Haijun Gong**, tBN-CSDI: A time-varying blue noise-based diffusion model for time series imputation. *Bioinformatics Advances* (2025)

A6. Yunge Wang**, Lingling Zhang, Tong Si, Sarah Roberts*, Yuqi Wang, **Haijun Gong**, Reconstructing Dynamic Gene Regulatory Networks Using f-Divergence From Time-Series scRNA-Seq Data. *Current Issues in Molecular Biology*, 47(6), (2025)

A7. Lingling Zhang, Yunge Wang**, Tong Si, Lucas Koch*, Sarah Roberts*, **Haijun Gong**, Time-varying Gene Regulatory Networks Inference Using KL Divergence From Single Cell Data. Proceedings of 17th International Conference on Bioinformatics and Biomedical Technology (2025)

A8. Wen-shan Liu**, Tong Si, Aldas Kriauciunas*, Snell M*, **Haijun Gong**. Bidirectional f-Divergence-Based Deep Generative Method for Imputing Missing Values in Time-Series Data. *Stats*. 8(1) (2025)

A9. Yunge Wang**, Lingling Zhang, Tong Si, Graham Bishop*, **Haijun Gong**. Anomaly Detection in High-Dimensional Time Series Data with Scaled Bregman Divergence. *Algorithms*. 18(2) (2025)

A10. Tong Si**, Yunge Wang**, Lingling Zhang, Evan Richmond*, Tae-Hyuk Ahn, **Haijun Gong**. Multivariate Time Series Change- Point Detection with a Novel Pearson-like Scaled Bregman Divergence. *Stats*;7(2):462-480 (2024)

A11. Tong Si**, Zackary Hopkins**, John Yanev*, Jie Hou, **Haijun Gong**, A novel f-divergence based generative adversarial imputation method for scRNA-seq data analysis. *PLoS One* 18(11): e0292792. (2023)

A12. Helen Richards*, Yunge Wang**, Tong Si**, Hao Zhang*, **Haijun Gong**, Intelligent Learning and Verification of Biological Networks. *Advances in Artificial Intelligence, Computation, and Data Science* (2021)

A13. W. Feng, Z. Yu, M. Kang, **Haijun Gong**, T.-H. Ahn, Practical Evaluation of Different Omics Data Integration Methods. Studies in Computational Intelligence, International Workshop on Health Intelligence (W3PHAI'19) In conjunction with the *33rd AAAI Conference on Artificial Intelligence* (AAAI-19) (2019)

A14. Zi Wang*, Yun Guo*, **Haijun Gong**, An Integrative Analysis of Time-varying Regulatory Networks from High-dimensional Data, *IEEE International Conference on Big Data*, Page 3798-3807 (2018)

A15. McCoy, M., Paul, A., Victor, M., Richner, M., Gabel, **Haijun Gong**, H., Yoo, A., Ahn, T. (2018), LONGO: an R package for interactive gene length dependent analysis for neuronal identity, *Bioinformatics*, V34, Issue 13: i422-i428.

A16. Ma, Y.**, Feng, L.*, Guo, Y., **Haijun Gong** (2016). Statistical Analysis and Probabilistic Verification of Stress-induced Signaling Pathways, *International Journal of Data Mining and Bioinformatics*, Vol. 14, No. 2.

A17. Ma, Y.**, Zhang, L, Geneus, C.**, **Haijun Gong** (2015). Weighted Gene Coexpression Network Analysis of Prostate Cancer, *IEEE International Conference on Computational Advances in Bio and Medical Sciences* (ICCABS).

A18. Ma, Y.**, Damazyn, K.*, Klinger, J.*, **Haijun Gong** (2015), Inference and Verification of Probabilistic Graphical Models from High-dimensional Data, International Conference on Data Integration in the Life Sciences, *Lecture Notes in Bioinformatics* (LNBI), Vol. 9162.

A19. **Haijun Gong**, Klinger, J.*, Damazyn, K.*, Li, X.**, Huang, S.* (2015). A novel procedure for statistical inference and verification of gene regulatory subnetwork, *BMC Bioinformatics*, V16, S7.

A20. **Haijun Gong**, Feng, L.* (2014). Computational Analysis of the Roles of ER-Golgi Network in the Cell Cycle. *BMC Systems Biology*, 8, S4.

A21. **Haijun Gong**, Feng, L.* (2014). Probabilistic Verification of ER Stress-induced Signaling Pathways. Proceedings of *IEEE International Conference on Bioinformatics and Biomedicine* (BIBM, UK).

A22. **Haijun Gong**, Wu, T.T., Clarke, E.M. (2014). Pathway-Gene Identification for Pancreatic Cancer Survival via Doubly Regularized Cox Regression. *BMC Systems Biology*, 8, S1:S3.

A23. **Haijun Gong**, Tong Tong Wu, Edmund M. Clarke (2014). Statistical Analysis of High-Dimensional Data for Pancreatic Cancer", Molecular Diagnostics and Treatment of Pancreatic Cancer: Systems and Network Biology Approaches (Chapter 6). Elsevier Academic Press.

A24. **Haijun Gong** (2013). Analysis of Intercellular Signal Transduction in the Tumor Microenvironment. *BMC Systems Biology*, 7, S3.

A25. **Haijun Gong**, Zuliani, P., Clarke, E.M. (2013) Model checking of a synchronous diabetes-cancer logical network. *Current Bioinformatics*, Volume 8: 9-15.

A26. **Haijun Gong**, Zuliani, P., A. Komuravell, Faeder, J., Clarke, E.M. (2012). Computational modeling and verification of signaling pathways in cancer. Algebraic and Numeric Biology, *Lecture Notes in Computer Science* (LNCS), Vol. 6479.

A27. Wu, T.T., **Haijun Gong**, Clarke, E.M. (2011). A Transcriptome analysis by Lasso penalized Cox regression for pancreatic cancer survival. *Journal of Bioinformatics and Computational Biology*, V 9.

A28. **Haijun Gong**, Zuliani, P., Wang, Q., Clarke, E.M. (2011). Formal analysis for logical models of pancreatic cancer. Proceedings of 50th *IEEE Conference on Decision & Control and European Control Conference*.

A29. **Haijun Gong**, Wang, Q., Zuliani, P., Faeder, J., M. Lotze, Clarke, E.M. (2011). Symbolic model checking of signaling pathways in pancreatic cancer. Proceedings of 3rd *International Conference on Bioinformatics and Computational Biology* (BICoB).

A30. **Haijun Gong**, P. Zuliani, E. Clarke, "Model Checking of a Diabetes-Cancer Model", International symposium on computational models for life sciences, AIP Conf. Proc. 1371 (2011)

A31. **Haijun Gong**, Zuliani, P., Komuravell, A., Faeder, J., Clarke, E.M. (2010). Analysis and verification of the HMGB1 signaling pathway. *BMC Bioinformatics*, 11 (S7).

A32. Ayorkor Korsah, G., Mostow, J., MB Dias, TM Sweet, S. Belousov, MF Dias, **Haijun Gong** (2010), "Improving Child Literacy in Africa: Experiments with an Automated Reading Tutor", Information Technologies & International Development.

A33. Ayorkor Mills-Tettey, G., J. Mostow, MB Dias, TM Sweet, S. Belousov, MF Dias, **Haijun Gong** (2009), "Improving Child Literacy in Africa: Experiments with an Automated Reading Tutor", International Conference on Information and Communication Technologies & Development.

A34. **Haijun Gong**, Guo, Y., Linstedt, A., Schwartz, R. (2010). Discrete, continuous and stochastic models of protein sorting in the Golgi. *Physical Review E*, 81.

A35. **Haijun Gong**, Sengupta, D., Linstedt, A., Schwartz, R. (2008). Simulated de novo assembly of Golgi compartments by selective cargo capture during vesicle budding and targeted vesicle fusion. *Biophysical Journal*, Volume 95.

A36. **Haijun Gong**, Sun, J., Du, D. (2002), "Analysis of Ds → Φ Л Beyond Naïve Factorization", High Energy Physics and Nuclear Physics, Volume 26.

A37. Du, D., **Haijun Gong**, Sun, J., Yang, D., Zhu, G. (2002), "Phenomenological Analysis of Charmless Decays B → PV with QCD Factorization", Physical Review D 65, 094025.

A38. Du, D., **Haijun Gong**, Sun, J., Yang, D., Zhu, G. (2002), "Phenomenological Analysis of B → PP Decays with QCD Factorization", Physical Review D 65, 074001.

Professional Activities

- NSF Proposal Review Panelist: 2015 – Present
- NIH Proposal Review Panelist: 2019
- Associate Editor – Heliyon Journal: 2023 – Present
- Alzheimer's Association International Research Program Grant Reviewer (2013)
- Program co-Chair, International Conference on Bioinformatics and Biomedical Technology (2021-2023)
- Local organizing committee: International Meeting on Information and Statistics in Nuclear Experiment and Theory (ISNET-9), Washington University St. Louis, May 22-26, 2023
- Computers in Biology and Medicine, Editorial Board Member 2013 - April 2015
- International Society for Computational Biology (ISCB): Member
- Review papers for the following journals:
  Nature Genetics; BMC Systems Biology; BMC Developmental Biology; Biometrics; PLOS Computational Biology; IEEE/ACM Transactions on Computational Biology and Bioinformatics; International Journal of Machine Learning and Cybernetics; Computers in Biology and Medicine; Journal of Theoretical Biology; Journal of Chemical Physics; Biophysical Journal; Mathematical Biosciences; SLU Master's Program in Bioinformatics and Computational Biology
- SLU Science and Engineering Research Council (SERC) Representative 2020 – 2025
- SLU CAS Academic Affairs Committee: 2014 – 2015
- Grant Reviewer for President's Research Funds (PRF)
- Judge of Annual Graduate Research Symposium

Awards

2010	Best Paper Award, International Conference on Bioinformatics (InCoB'10)
2009	"Honorable Mention" Student Paper Award, ICICTD conference

Teachings at SLU

1. Probability Theory: 2012 Fall, 2013 Fall, 2015 Spring, 2019 Fall, 2021 Fall
2. Mathematical Statistics: 2013 Spring, 2015 Fall, 2016 Spring, 2020 Spring
3. Applied Regression Analysis: 2014 Spring, 2016 Spring, 2018 Spring, 2023 Spring, 2024 Spring, 2025
4. Time Series Data Analysis: 2016 Fall, 2017 Fall, 2022/2023 Spring
5. Bayesian Data Analysis and Statistical Computing: 2017 Spring, 2022/2023 Fall
6. Statistical Models: 2024 Spring
7. Elementary Statistics with Computers: 2014 Spring
8. Probability & Statistics for Engineers: 2012 Fall, 2013 (Spring & Fall), 2014 Spring, 2015 (Spring, Fall)
9. Foundations of Statistical Analysis: 2016 Fall, 2017 (Spring & Fall), 2018 Spring, 2019-2022

Teaching at Australian National University

Applied Statistics: 2019 Winter

Students Mentoring at SLU

I have supervised 2 PhD students, 11 Master students, over 20 undergraduate students on statistics and bioinformatics research since 2012.